

(20519)

Roll No. 3508

Total Questions : 10]

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3508

B.Sc. (Microbiology) IIIrd Year Examination,
May-2019

RECOMBINANT DNA TECHNOLOGY

(B-305)

[B.Sc. (Micro.)]

Time : 3 Hrs.]

[M.M. : 50

Note :- Attempt any *five* questions. All questions carry equal marks.

1. Describe in detail the procedure of gene cloning and its uses.

10

2. Describe the contributions of any *four* :

(i) Temin and Baltimore

(ii) Jackson, Symons and Berg

NA-339

(1)

Turn Over

(iii) Southern

(iv) Mullis

(v) Arber and Smith

$2\frac{1}{2} \times 4 = 10$

3. Write short notes on the following :

(i) Restriction endonucleases

(ii) Cosmids

(iii) BAC vectors

(iv) YAC vectors

$2\frac{1}{2} \times 4 = 10$

4. Describe the procedure of construction of c-DNA library. 10

5. Describe in detail procedure of DNA sequencing given by Sanger. 10

6. How will you identify expression of foreign gene in prokaryotes ? Describe in detail. 10

7. Write notes on the following :

(i) Northern blotting

(ii) Western blotting

$5 \times 2 = 10$

8. Describe in detail about various types of PCR and their applications. 10

9. Describe the process of monoclonal antibodies production.

10

10. Write short notes on the following :

(i) Hepatitis B vaccines

(ii) Insulin

(iii) DNA probes

(iv) Construction of chimeric DNA

$2^{1/2} \times 4 = 10$

A

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(20620)

Roll No.

B.Sc. (Micro) - III Yr.

3508 (CV)

B.Sc. (Microbiology) Examination,

June - 2020

Recombinant DNA Technology

(B-305)

Time : Two Hours]

[Maximum Marks : 50

Note : Attempt any **four** questions. **All** questions carry equal marks.

1. Describe the technique of Southern, Western and Northern blotting. Describe the applicability of these techniques in biotechnological work.
2. What do you understand by recombinant DNA technology? Discuss the various applications in biotechnological

P.T.O.

programmes.

3. Write detailed note on :
 - (a) Restriction enzyme
 - (b) Monoclonal antibodies
4. What is PCR? Discuss various applications of PCR in biotechnological research.
5. How can DNA probes be prepared and labelled? Discuss their use in biotechnological research.
6. Write note on :
 - (a) Production of human growth hormone
 - (b) Site directed mutagenesis
7. Explain the requirements of recombinant molecules in medical, agriculture and industrial sectors.
8. Write short note on the following :
 - (a) cDNA library

(b) Insulin

(c) Plasmid

(d) BAC

9. Describe different kinds of vectors used for gene cloning.

10. Compare the Sanger's and Maxam Gilbert's method of gene sequencing.

D

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(20721)

Roll No. ...180979131021

B.Sc. (Micro.)-III Yr.

3508 (CV-II)

B.Sc. (Microbiology) Examination,

June - 2021

RECOMBINANT DNA TECHNOLOGY

(B-305)

[B.Sc. (Micro.)]

Time : 1½ Hours]

[Maximum Marks : 50

Note : Attempt any **two** questions. **All** questions carry equal marks.

1. What is meant by sequencing of nucleic acids. Discuss the Sanger's method of DNA sequencing in detail.
2. What do you understand by recombinant DNA technology? Discuss the various applications of this technology in biotechnology.

P.T.O.

3. Write detailed note on the following :
- (a) Site directed mutagenesis
 - (b) Hepatitis B Vaccines.
4. What are genomic and cDNA libraries?
How can these be prepared? Discuss their utility in biotechnology programme.
5. Write short note on the following :
- (a) Restriction enzymes
 - (b) Shotgun experiment
 - (c) Western blotting
6. How can molecular probe be prepared and labelled? Discuss their uses in biotechnology.
7. Differentiate between the following :
- (a) Cloning vector and expression vector

(b) Southern blotting and Western blotting

(c) Chromosome walking and chromosome jumping

(d) RFLPs and RAPDs

8. What are the features of a good vector?

Describe the features of one plasmid and one phage vector of **E.coli**.

9. What is PCR? Discuss the various steps of PCR and its applications in biological research.

10. What is gene cloning? Discuss the various methods of gene cloning.

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(20622)

Roll No.

B.Sc.(Micro.) - III Yr.

3508

**B.Sc. (Microbiology) Examination,
June-2022**

RECOMBINANT DNA TECHNOLOGY

(B-305)

[B.Sc.(Micro.)]

Time : Three Hours / [Maximum Marks : 50]

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. What is c-DNA? Discuss the various steps involved in construction of c-DNA library. 10
2. Write the detailed notes on the following: 5 each
 - (a) Restriction endonucleases
 - (b) Site directed mutagenesis

P.T.O.

3. Explain some characteristic features and process of Southern blotting. How it differ from Northern blotting. 10
4. Write in detail the important contribution made in the field of biotechnology of any two : 5 each
- (a) Temin and Baltimore
 - (b) Arber and Smith
 - (c) Mullis
5. What is PCR? Describe the different variants of PCR. Discuss the utility with examples. 10
6. What do you understand by recombinant DNA technology? Discuss the applicability in biotechnological programmes. 10
7. Write short notes on the following : 2½ each
- (a) Chromosome walking

(b) Gene amplification

(c) YAC vectors

(d) Monoclonal antibodies

8. What are the properties of a good vector?

Discuss the different kinds of vectors used for gene cloning. 10

9. Discuss the Sanger's method of gene sequencing in detail. 10

10. Write detailed note on the following :

5 each

(a) Hepatitis B vaccines

(b) Expression of foreign genes in prokaryotes

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(20524)

Roll No.

B.Sc. (Micro.)-III Year

3508

B.Sc. (Microbiology)

Examination, May-2024

RECOMBINANT DNA TECHNOLOGY

(B-305)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All**
questions carry equal marks.

1. Explain the characteristic features and process of Southern blotting. How does it differ from Northern blotting. 10
2. Explain the production of human growth hormones and monoclonal antibodies. 10

P.T.O.

3. Differentiate between :

5 each

(a) Animal and Plant viral vectors.

(b) cDNA library and genomic library.

4. Discuss the contributions of : 2½ each

(a) Arber and Smith.

(b) Mullis.

(c) Temin and Baltimore.

(d) Jackson, Symons and Berg.

5. Describe in detail the procedure of gene cloning and its uses. 10

6. Describe in detail about various types of PCR and their applications. 10

7. Discuss the various differences in expression of foreign genes in Prokaryotes and eukaryotes. 10

8. What is the principle of recombinant DNA technology? Discuss the various steps involved in this process. 10

9. Write short note on the following : 2½ each

(a) Site directed mutagenesis.

(b) Monoclonal antibodies.

(c) Hepatitis b vaccines.

(d) DNA probes.

10. Discuss the procedure of DNA sequencing given by Sanger in detail. 10

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(20425)

Roll No.12.

B.Sc. (Micro.)-III Year

3508

B.Sc. (Microbiology)

Examination, April-2025

RECOMBINANT DNA TECHNOLOGY

(B-305)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. Describe any two types of restriction enzymes, including examples and their recognition sites?
2. Write detailed note on the following (any **two**) :
 - (i) Site directed mutagenesis
 - (ii) Monoclonal antibodies

P.T.O.

- (iii) Gene cloning and its applications
3. Discuss the applications of recombinant technology in agriculture?
 4. Write short note on the following :
 - (i) Plasmid
 - (ii) Cosmid
 - (iii) Bacteriophages
 - (iv) cDNA
 5. Explain how cDNA libraries can be used in gene expression studies and the advantages they offer over genomic libraries?
 6. Give a comparative study on foreign gene expression in prokaryotes & eukaryotes.
 7. Write detailed note on the following :
 - (i) PCR & its application
 - (ii) Expression vectors

8. Differentiate between Southern, Northern and Western blotting techniques.
9. Describe Human growth hormone. Explain any two of them with its production and significance?
10. What is hybridisation technique? Explain the concept of fluorescent in situ hybridisation (FISH).